

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA IIT - JEE ACADEMY - INDIA

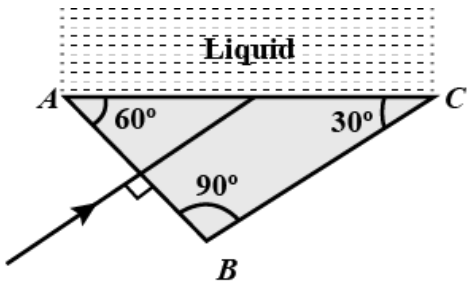
Sec: SR.IIT_*CO-SC(MODEL-A& APEX) **GTM-23** **Date: 22.01.2025**
Time: 3.00Hrs. **JEE MAIN MODEL** **Max. Marks: 300**

PHYSICS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

01. Light is incident normally of face AB of a prism as shown in figure. A liquid of refractive index μ is placed on face AC of the prism. The prism is made of glass of refractive index $3/2$.



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The limit of μ for which total internal reflection takes placed on face AC is

A) $\mu > \frac{3}{4}$ B) $\mu < \frac{3\sqrt{3}}{4}$ C) $\mu < \sqrt{3}$ D) $\mu < \frac{\sqrt{3}}{2}$

02. Three concentric metallic spherical shells of radii R, 2R, 3R are given charges Q_1, Q_2, Q_3, respectively. It is found that the surface charge densities on the outer surface of the shells are equal. Then, the ratio of the charges given to the shells $Q_1 : Q_2 : Q_3$ is

A) 1 : 8 : 18 B) 1 : 3 : 5 C) 3 : 4 : 5 D) 1 : 4 : 9

03. A coil of inductive reactance 31Ω has a resistance of 8Ω. It is placed in series with a condenser of capacitive reactance $25\,\Omega$. The combination is connected to an AC source of 110V. The power factor of the circuit is

A) 0.56 B) 0.64 C) 0.80 D) 0.33

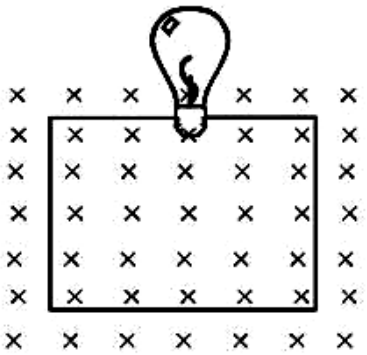
04. Consider a rocket is being fired. The kinetic energy of the rocket is increased by 16 times where as its total mass is reduced by half through the burning of fuel. The factor by which its momentum increases is

A) 8 B) $2\sqrt{2}$ C) 4 D) $4\sqrt{2}$

05. The work functions of metals A and B are in the ratio 1 : 2. If light of frequencies ' f ' and ' 2f ' are incident on the surfaces of A and B respectively, the ratio of the maximum kinetic energies of photoelectrons emitted is (f is greater than threshold frequency of A, ' 2f ' is greater than threshold frequency of B)

A) 1 : 1 B) 1 : 2 C) 1 : 3 D) 1 : 4

06. A square wire loop of 10.0 cm side lies at right angle to a uniform magnetic field of 20T. A 10 V light bulb is in a series with the loop as shown in the fig. The magnetic field is decreasing steadily to zero over a time interval Δt. The bulb will shine with full brightness if Δt equal to



A) 20 ms B) 0.02 ms C) 2 ms D) 0.2 ms

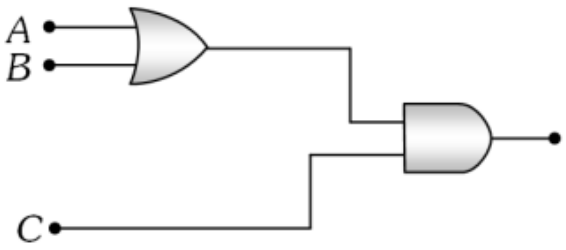
07. An organ pipe P_1, closed at one end and vibrating in its first overtone, and another pipe P_2 open at both ends and vibrating in its third overtone, are in resonance with a given tuning fork. The ratio of the length of P_1 to that of P_2 is (end corrections are negligible)

A) $\frac{8}{3}$ B) $\frac{3}{8}$ C) $\frac{1}{2}$ D) $\frac{1}{3}$

08. Two wires of the same metal have same length, but their area of cross – sections are in the ratio 3 : 1. They are joined in series. The resistance of thicker wire is $10\,\Omega$. The total resistance of the combination will be

A) $20\,\Omega$ B) $10\,\Omega$ C) $100\,\Omega$ D) $40\,\Omega$

09. To get an output 1 from the circuit shown in the figure, the inputs must be



A) A = 0, B = 1, C = 0 B) A = 1, B = 0, C = 0

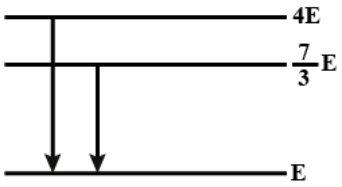
C) A = 1, B = 0, C = 1 D) A = 1, B = 1, C = 0

10. A plane electromagnetic wave of frequency 100 MHz is travelling in vacuum along the x - direction. At a particular point in space and time $\vec{B}=2.0\times10^{-8}\hat{k}T$. (Where, $\hat{k}$ is unit vector along z-direction) What is $\vec{E}$ at this point? (speed of light $c=3\times10^{8}m/s$)

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A) $6.0\hat{k}\,V/m$ B) $0.6\hat{k}\,V/m$ C) $6.0\ \hat{j}\,V/m$ D) $0.6\ \hat{j}\,V/m$

11. The following diagram indicates the energy levels of a certain atom when the system moves from $4E$ level to E, a photon of wavelength λ_1 is emitted. The wavelength of photon produced during it's transition from $\frac{7}{3}E$ level to E is λ_2. The ratio $\frac{\lambda_1}{\lambda_2}$ will be



A) $\frac{9}{4}$ B) $\frac{4}{9}$ C) $\frac{3}{2}$ D) $\frac{7}{3}$

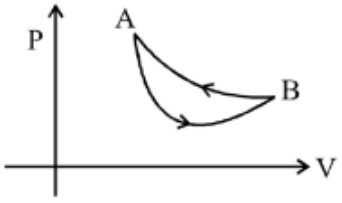
12. v_e and v_p denotes the escape velocity from the earth and another planet having twice the radius and the same mean density as the earth. Then

A) $v_e = v_p$ B) $v_e = v_p/2$ C) $v_e = 2v_p$ D) $v_e = v_p/4$

13. The magnetic susceptibility of a material of a rod is 499. Permeability of vacuum is $4\pi\times10^{-7}\,Hm^{-1}$. Absolute permeability of the material of the rod (in Hm^{-1}) is

A) $\pi\times10^{-10}$ B) $2\pi\times10^{-4}$ C) $3\pi\times10^{-4}$ D) $4\pi\times10^{-4}$

14. A gas is undergoing a cyclic process as shown in figure. For this cyclic process mark out the correct option.



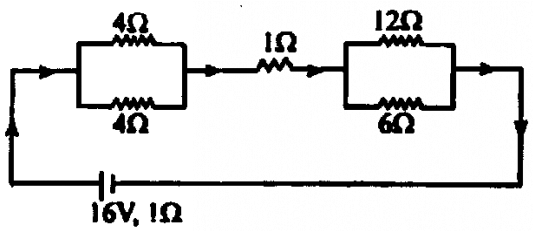
A) Heat is rejected from the system

B) Work is being done by the system on the surrounding.

C) Net heat is supplied to system in full cycle.

D) In full cycle internal energy of system is increasing.

15. A network of resistors is connected to a 16V battery with internal resistance of 1Ω as shown. The current in the 6Ω resistor is



A) 1A B) 1.33 A C) 2 A D) 2.5 A

16. A proton and an α-particle are accelerated through the same potential difference. The ratio of de-Broglie wavelength of the proton to that of the α particle will be

A) 2:1 B) 1:1 C) 1:2 D) $2\sqrt{2}:1$

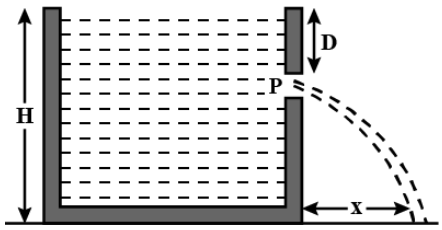
17. The dimensional formula for the physical quantity $\frac{B\mu_0\varepsilon_0}{E}$ is :- (here E = intensity of electric field, B = magnetic induction and symbols have their usual meaning)

A) $L^0M^0T^0$ B) $L^1M^0T^{-1}$ C) $L^{-3}M^0T^3$ D) $L^{1/2}M^0T^{-1/2}$

18. An air bubble of radius 0.1 cm is in a liquid having surface tension 0.06 N/m and density $10^3\,kg/m^3$. The pressure inside the bubble is $1100\,Nm^{-2}$ greater than the atmospheric pressure, At what depth is the bubble below the surface of the liquid $\left(g=9.8ms^{-2}\right)$

A) 0.1m B) 0.15m C) 0.20m D) 0.25m

19. A tank is filled with water up to a height H. Water is allowed to come out of a hole P in one of the walls at a depth D below the surface of water. Express the horizontal distance x in terms of H and D



A) $x=\sqrt{D(H-D)}$ B) $x=\sqrt{\frac{D(H-D)}{2}}$

C) $x=2\sqrt{D(H-D)}$ D) $x=4\sqrt{D(H-D)}$

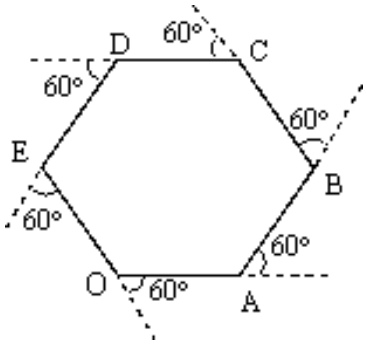
20. A spaceship is moving with speed v_0 in gravity free space along +y axis suddenly shoots out one third of its part with speed $2v_0$ along +x axis. The speed of remaining part is

A) $2v_0$ B) $\frac{v_0}{2}$ C) $\frac{\sqrt{13}}{2}v_0$ D) $4v_0$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

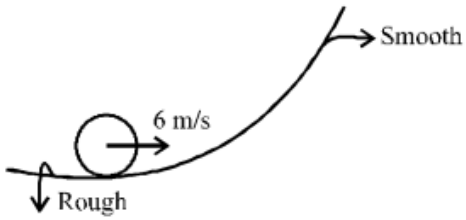
21. A person moving on a motor cycle in a ground takes a turn through 60^0 on his left after every 50m. Then find the magnitude of displacement suffered by him after 9th turn is ____m



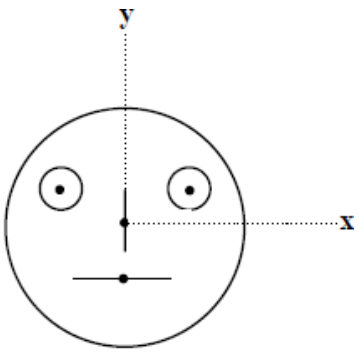
22. In a Young's double slit experiment using monochromatic light of wavelengths λ, the intensity of light at a point on the screen with a path difference λ is M units and the intensity of light at a point where the path difference is $\frac{\lambda}{3}$ in n M units. What is the value of 100 n? (intensity of light waves are equal)

23. A heavy nucleus mass number 200 gets disintegrated into smaller fragments of mass number 80 and 120. If binding energy per nucleon for parent nucleus is 6.5 MeV and for daughter nuclei are 7 MeV and 8 MeV respectively and the energy released in the decay is given by $X\ MeV$, then X will be

24. A disc of radius 0.1 m rolls without sliding on a horizontal surface with a velocity of 6 m/s. It then ascends a smooth continuous track as shown in figure. The height upto which it will ascend is $\frac{X}{5}$ m. Value of x is: - [Take g = 10 m/s^2]



25. Look at the drawing given in the figure which has been drawn with ink of uniform line-thickness. The mass of ink used to draw each of the two inner circles, and each of the two-line segment is m. The mass of the ink used to draw the outer circle is $6m$. The coordinates of the centres of the different parts are: outer circle $(0,0)$, left inner circle $(-a,a)$, right inner circle (a,a), vertical line $(0,0)$ and horizontal line $(0,-a)$. The y-coordinate of the centre of mass of the ink in this drawing is $\frac{a}{n}$ then value of n is__



CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Given below are the two statements:

Statement-I: Noble gases have large positive values of electron gain enthalpy.

Statement-II: Noble gases have stable electronic configurations, they have no tendency to accept the electron, hence have large positive values of electron gain enthalpy**.**

In the light of above statements, choose the **correct answer** from the options given below:

A) Statement I is false but Statement II is true

B) Both Statement I and Statement II are true

C) Statement I is true but Statement II is false

D) Both Statement I and Statement II are false

27. Consider the reaction in aqueous medium, which is in red colour

$$\underset{\text{pale yellow}}{Fe^{3+}(aq)} + 3\underset{\text{colourless}}{SCN^{-}(aq)} \rightleftharpoons \underset{\text{deep red}}{Fe(SCN)_3}$$

What happens if we add aqueous $HgCl_2$ to the above equilibrium?

A) Intensity of red colour increases

B) Solution turns blue due to the formation of $Hg(SCN)_2$

C) Concentration of Fe^{3+} ion decreases

D) Intensity of red colour decreases

28. When acetone and chloroform are mixed together, H-bonding takes place between them. Such a liquid pair show

A) Positive deviation from Raoult's law B) Negative deviation from Raoult's law

C) No deviation from Raoult's law D) Slight increase in volume

29. Correct combination of compound and colour is-

A) $Anhydrous\,CuSO_4\,crystals---Blue$

B) $KMnO_4\,crystals---Orange$

C) $FeSO_4.7H_2O\,crystals---Pale\,green$

D) $AgCl\,crystals---yellow$

30. Some statements regarding batteries are given below

I) In dry cell, at cathode, oxidation state of Mn changes from +4 to +3

II) In Mercury cell, at cathode, oxidation state of Hg changes from +2 to zero.

III) In both dry cell and Mercury cell at anode, Zn undergoes oxidation.

IV**)** In nickel-cadmium cell, at cathode, oxidation state of Ni changes from +3 to +2

Choose the correct answer from the options given below:

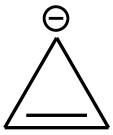
A) I, II and III only

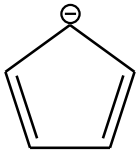
B) I, II, III and IV

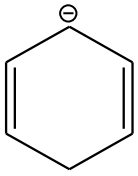
C) II and IV only

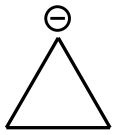
D) I and III only

31. Which of the following carbanion is aromatic?

A) 

B) 

C) 

D) 

32. Calculate the equivalent conductivity of $1M\,H_2SO_4$ solution when its specific conductance is equal to $26\times10^{-2}\,S\,cm^{-1}$.

A) $1.3\times10^{2}\,S\,cm^{2}eq^{-1}$

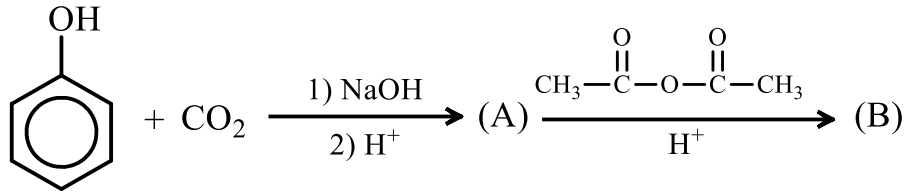
B) $1.6\times10^{2}\,S\,cm^{1}$

C) $13\,S\,cm^{2}mol^{-1}$

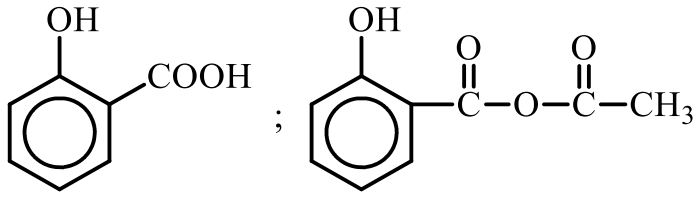
D) $1.3\times10^{3}\,S\,cm^{2}mol^{-1}$

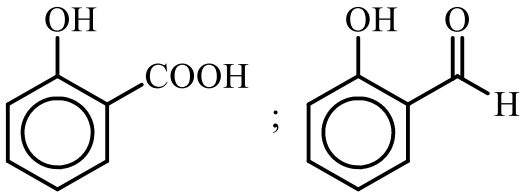
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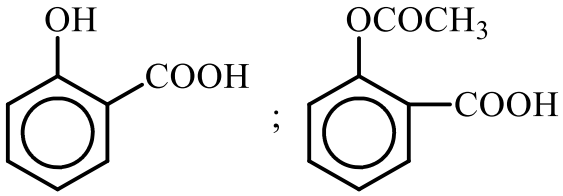
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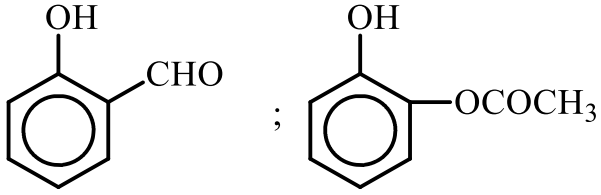


Product (A) and (B) respectively are

A) 

B) 

C) 

D) 

34. Given below are the two statements: One is labelled as **Assertion A** and the other labelled as **Reason R:**

Assertion A: Acetylation of alcohols with acid anhydride and acid chloride is carried out in presence of small amount of concentrated sulphuric acid.

Reason R: Acetylation of alcohols with acid anhydride and acid chloride is a reversible reaction.

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Correct answer from the options given below:

A) A is false but R is true

B) A is true but R is false

C) Both A and R are true and R is the correct explanation of A

D) Both A and R are true and R is NOT the correct explanation of A

35. The increasing order of ionic radii of the given isoelectronic species is:

I) S^{2-} II) Cl^{-} III) K^{+} IV) Ca^{2+}

Choose the most appropriate from the options given below

A) $I < II < III < IV$ B) $II < III < I < IV$

C) $IV < III < I < II$ D) $IV < III < II < I$

36. Incorrect statement regarding $K_2Cr_2O_7$ among the following is_____

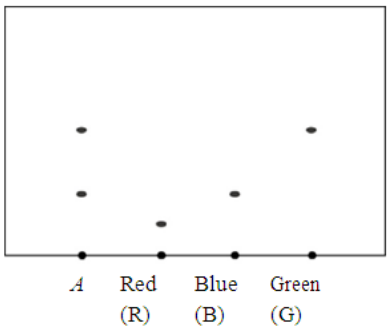
A) It reacts with aqueous NaCl in presence of Conc.H_2SO_4 to give chromyl chloride vapours.

B) Its intense colour is due to charge transfer.

C) It oxidizes H_2S to S in acidic medium.

D) As pH is increased in its aqueous solutions, it gets increasingly converted into chromate ion.

37. Four coloured substances were spotted on a strip of chromatography paper, which was then suspended in an alcohol-water mixture. The four coloured substances include dye *A*, and three basic dyes red, blue and green. After some time, the following chromatogram was obtained.



What colours does dye A contains?

A) (R) and (B) only B) (R) and (G) only

C) (B) and (G) only D) (R), (B) and (G)

38. The reaction $2NO + Br_2 \rightarrow 2NOBr$, obeys the following Mechanism

$NO + Br_2 \overset{fast}{\rightleftharpoons} NOBr_2$

$NOBr_2 + NO \xrightarrow{slow} 2NOBr$

The rate expression can be written as

A) $r = K[NO]^2[Br_2]$ B) $r = K[NO][Br_2]$

C) $r = K[NO][Br_2]^2$ D) $r = K[NOBr_2]$

39. Given below are the two statements: One is labelled as **Assertion A** and the other labelled as **Reason R:**

Assertion A: Benzenamine is weak base than Phenylmethanamine

Reason R: pK_a value of Benzenamine is 9.38 while that of Phenylmethanamine is 4.70

In the light of above statements, choose the **correct answer** from the options given below:

A) A is false but R is true

B) A is true but R is false

C) Both A and R are true and R is the correct explanation of A

D) Both A and R are true and R is NOT the correct explanation of A

40. Given below are the two statements:

Statement-I: Methoxy Benzene is commonly called as Anisole

Statement-II: Anisole is an ether

In the light of above statements, choose the most **appropriate answer** from the options given below:

A) Statement I is false but Statement II is true

B) Both Statement I and Statement II are true

C) Statement I is true but Statement II is false

D) Both Statement I and Statement II are false

41. The correct statements from the following are:

I) $[Ni(CO)_4]$ – Tetrahedral, paramagnetic

II) $\left[Ni(CN)_4\right]^{2-}$ – Square planar, diamagnetic

III) $\left[NiCl_4\right]^{2-}$ – Tetrahedral, paramagnetic

IV) $\left[Co(en)_3\right]^{3+}$ – Octahedral, optically active

Choose the correct answer from the options given below

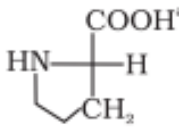
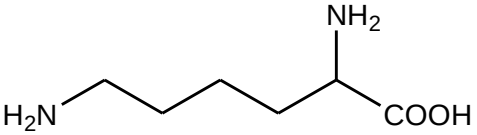
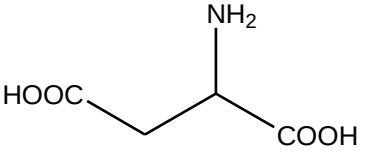
A) I, II only B) I, II, III only

C) II, III, IV only D) I, II, III and IV

42. Which of the following molecule has only two $\pi - bonds$ and is diamagnetic?

A) Li_2 B) C_2 C) N_2 D) O_2

43. Match List I (property) with List II (compound)

LIST-I		**LIST-II**	
P.	Achiral amino acid	I.	
Q.	Amino acid with secondary amine group	II.	
R.	Acidic amino acid	III.	H_2N-CH_2-COOH
S.	Basic amino acid	IV.	

Choose the correct answer from the options given below:

A) $P-III, Q-I, R-IV, S-II$ B) $P-II, Q-I, R-IV, S-III$

C) $P-I, Q-II, R-IV, S-III$ D) $P-III, Q-II, R-IV, S-I$

44. Which of the following is/are correct for group 14 elements?

(P) The stability of dihalides are in the order $CX_2 < SiX_2 < GeX_2 < SnX_2 < PbX_2$

(Q) The ability to form $p\pi - p\pi$ multiple bonds among themselves increases down the group

(R) The tendency for catenation decreases down the group

(S) They all form oxides with the formula MO_2

A) P, Q only B) P, Q, R only C) P, R, S only D) P, Q, R and S

45. Match List I (Extra element in the organic compound) with List II (Coloured compound formed in the identification test)

LIST-I		**LIST-II**	
P.	Nitrogen	I.	Violet coloured complex
Q.	Sulfur	II.	Blue coloured complex
R.	Phosphorus	III.	White coloured precipitate
S.	Chlorine	IV.	Yellow coloured precipitate

Choose the correct answer from the options given below:

A) $P-II, Q-I, R-III, S-IV$ B) $P-II, Q-I, R-IV, S-III$

C) $P-I, Q-II, R-IV, S-III$ D) $P-III, Q-II, R-IV, S-I$

APEX JEE-MAIN 2025

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

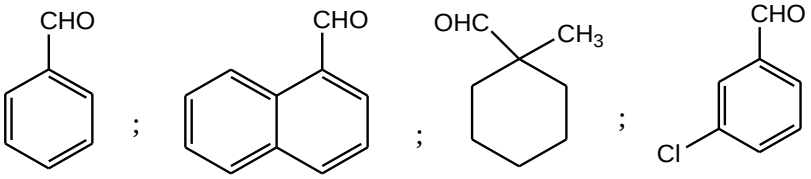
This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. Brown ring test is the conformation test for nitrate ion in qualitative analysis. Brown ring is obtained due to the formation of a complex compound. Magnitude of charge present on the complex ion is-

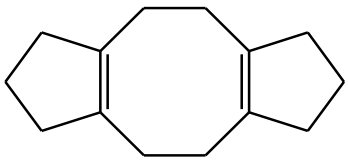
47. Total crystal field splitting energy (CFSE) for $\left[CoCl_6\right]^{4-}$ is $18000\,cm^{-1}$. Then CFSE for $\left[CoCl_4\right]^{2-}$ will be nearly $x\times10^3 cm^{-1}$.Find the value of x?

48. How many of the following compound(s) can undergo Cannizaro reaction?

$HCHO,\ CH_3CHO,\ CCl_3CHO,\ OHC-CHO$



49.



$\xrightarrow[2.H_2O/Zn]{1.O_3}$ Major product

Maximum number of sp^2-carbons present in the product is____

50. The value of equilibrium constant for the reaction $A \rightleftharpoons B$ at 298 K_{eq} is 10^x .'x'is____

(Given $\Delta H^0 298K = -54.07 KJ\,mol^{-1}$; $\Delta S^0 298K = 10\,JK^{-1}mol^{-1}$;

$R = 8.314\,JK^{-1}mol^{-1}$ *and* $2.303\times8.314\times298 = 5705$)

APEX JEE-MAIN 2025

MATHS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Two distinct complex number z_1 & z_2 satisfy the equation $z+1+8i=|z|(1+i)$, then $|z_1|+|z_2|=$_____

A) 16 B) 18 C) 13 D) $8\sqrt{2}$

52. **Statement – (S_1):** The equation $(x-a)(x-c)+\lambda(x-b)(x-d)=0$ where a<b<c<d has non real roots if $\lambda>0$.

Statement – (S_2): The range of $f(x)=(K-3)x^2-2Kx+3K-6$ is $[0,\infty)$(where $x\in R$), then $K=6,\frac{3}{2}$

A) S_1, S_2 both true B) S_1 is true S_2 is false
C) S_1, S_2 both false D) S_1 is false S_2 is true

53. If $\int\frac{\cos x-\sin x+1-x}{e^x+\sin x+x}dx=\ln\left[f(x)\right]+g(x)+C$ where C is the constant of integration and $f(x)$ is positive, then $f(x)+g(x)$ has the value equal to

A) $e^x+\sin x+2x$ B) $e^x+\sin x$
C) $e^x-\sin x$ D) $e^x+\sin x+x$

54. The length of the chord cut by the circle $x^2+y^2=2$ on the line $y-2x-1=0,$ is :

A) 0 B) $\frac{2}{9}$ C) $\frac{6}{\sqrt{5}}$ D) $\frac{4}{9}$

55. Consider two lines in space as $L_1:\vec{r_1}=\hat{j}+2\hat{k}+\lambda\left(3\hat{i}-\hat{j}-\hat{k}\right)$ and $L_2:\vec{r_2}=4\hat{i}+3\hat{j}+6\hat{k}+\mu\left(\hat{i}+2\hat{k}\right)$. If the shortest distance between these lines is $\sqrt{d}$ then d equals:

A) 5 B) 6 C) 7 D) 8

56. Words (with or without meaning) of 8 letters are formed using the letters of the word EQUATION. If the number of such words where exactly one letter is repeated twice and NO other letter is repeated, is k.8! then k =

A) 38 B) 28 C) 20 D) 22

57. Let the point (p, p+1) lie in the region $E=\left\{(x,y):3-x\le y<\sqrt{9-x^2},0\le x\le 3\right\}$. If the set of values of p is the interval [a,b), then b^2+b+a^2 is

A) 5 B) 4 C) 3 D) 6

58. The term independent of x in the expansion of $\left(1-x^2+3x^3\right)\left(\frac{5}{2}x^3-\frac{1}{5x^2}\right)^{11}, x\neq 0$ is

A) $\frac{7}{40}$ B) $\frac{33}{200}$ C) $\frac{39}{200}$ D) $\frac{11}{50}$

59. Let $f(x)=e^{|x^2-4x+3|}$ then :

A) f(x) decreases in the interval $(1,2)\cup(3,\infty)$

B) f(x) increases in the interval $(-\infty,1)\cup(2,3)$

C) f(x) has one local maximum point and two local minimum points

D) f(x) has one local minimum point and two local maximum points

60. Let $x=x(y)$ be the solution of the differential equation $2ye^{x/y^2}dx+\left(y^2-4xe^{x/y^2}\right)dy=0$ such that $x(1)=0$, then $x(e)$ is equal to

A) $e\log_e(2)$ B) $-e\log_e(2)$ C) $e^2\log_e(2)$ D) $-e^2\log_e(2)$

61. Let e and l respectively be the eccentricity and length of the latus rectum of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. Let e' and l' respectively be the eccentricity and length of the latus rectum of its conjugate hyperbola. If $e^2=\frac{11}{14}l$ and $(e')^2=\frac{11}{8}l'$, then the value of

$77a+44b$ is equal to $(a > 0,\ b>0)$

A) 100 B) 110 C) 120 D) 130

62. If the probability that a number selected from the set $\{1,2,3,.....,1000\}$ is divisible by 3 but neither divisible by 5 nor by 7 is $\frac{m}{n}$ (Where m and n are relatively prime numbers) , then (n-m) is

A) 667 B) 717 C) 771 D) 766

63. If $\begin{vmatrix} a^2+1 & ab & ac \\ ba & b^2+1 & bc \\ ca & cb & c^2+1 \end{vmatrix} = k\sqrt{(abc)},$ where a, b, c are positive real then minimum possible value of k is :

A) 1 B) 4 C) 16 D) 64

64. If 'a' be the digit at unit's place in $11^{2012}+23^{2012}-3^{2012}$, then $\int_{a-1}^{a} \frac{dx}{\sqrt{1-x^2}-x+\frac{1}{x}} =$

A) $\frac{\pi}{6}$ B) $\frac{\pi}{3}$ C) $\frac{\pi}{2}$ D) $\frac{\pi}{4}$

65. Let $H_n = 1+\frac{1}{2}+.....+\frac{1}{n}$, then the sum to n terms of the series $\frac{1^2}{1^3}+\frac{1^2+2^2}{1^3+2^3}+\frac{1^2+2^2+3^2}{1^3+2^3+3^3}+.....$ is

A) $\frac{4}{3}H_n-1$ B) $\frac{4}{3}H_n+\frac{1}{n}$ C) $\frac{4}{3}H_n$ D) $\frac{4}{3}H_n-\frac{2}{3}\frac{n}{n+1}$

66. If the system of equations $x+2y+3z=3,\ 4x+3y-4z=4,\ 8x+4y-\lambda z=9+\mu$ has infinitely many solutions, then the ordered pair (λ,μ) is equal to

A) $\left(\frac{72}{5},-\frac{69}{5}\right)$ B) $\left(-\frac{72}{5},\frac{21}{5}\right)$ C) $\left(\frac{72}{5},-\frac{21}{5}\right)$ D) $\left(\frac{72}{5},\frac{21}{5}\right)$

67. The complete set of values of x such that $\left[\cos^{-1}x\right]+\left[\cot^{-1}x\right]=0$, where [.] denotes greatest integer function, is

A) $(\cos 1,1]$ B) $(\cos 1,2\cos 1)$ C) $(\cot 1,1]$ D) $(0,\cot 1)$

68. The equations of the sides AB, BC and CA of a triangle ABC are: $2x+y=0,\ x+py=21a,(a\neq 0)$ and $x-y=3$ respectively. Let $P(2,a)$ be the centroid of ΔABC ,then $(BC)^2$ is equal to

A) 120 B) 122 C) 121 D) 141

69. Let $P(x,y)$ be a variable point on the curve $4x^2+9y^2-8x-36y+15=0$, then $\min\left(x^2-2x+y^2-4y+5\right)+\max\left(x^2-2x+y^2-4y+5\right)$ is equal to

A) $\frac{36}{325}$ B) $\frac{25}{13}$ C) $\frac{13}{25}$ D) $\frac{325}{36}$

70. Consider three observation a, b and c such that b = a+ c. If the standard deviation of a + 2, b + 2, c + 2 is d, then which of the following is true?

A) $b^2=3\left(a^2+c^2+d^2\right)$ B) $b^2=a^2+c^2+3d^2$

C) $b^2=3\left(a^2+c^2\right)-9d^2$ D) $b^2=3\left(a^2+c^2\right)+9d^2$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. If α, β are the roots of the quadratic equation $ax^2+bx+c=0$ then

$\lim\limits_{x\to\alpha}\dfrac{1-\cos\left(ax^2+bx+c\right)}{\left(x-\alpha\right)^2}$ equal to $\dfrac{a^2}{k}\left(\alpha-\beta\right)^2$ then k = _____

72. Let f be a differentiable function defined on $\left[0,\dfrac{\pi}{2}\right]$ such that $f\left(x\right)>0$ and

$f\left(x\right)+\int\limits_0^x f\left(t\right)\sqrt{1-\left(\log_e f\left(t\right)\right)^2}\,dt=e, \forall x\in\left[0,\dfrac{\pi}{2}\right]$, then $\left(6\log_e f\left(\dfrac{\pi}{6}\right)\right)^2$ is equal to

73. Let $A=\{1,2,3,4,5\}$ and $R=\{(1,2),(2,3),(1,4)\}$ be a relation on A. Let S be the equivalence relation on A such that $R\subset S$ and the number of elements in S is n, then, the minimum value of n is _____

74. If $\vec{a}$ *and* $\vec{b}$ are two vectors such that $\left|\vec{a}\right|=1, \left|\vec{b}\right|=4,\ \ \vec{a}.\vec{b}=2$. If $\vec{c}=\left(2\vec{a}\times\vec{b}\right)-3\vec{b}$, then angle between $\vec{b}$ *and* $\vec{c}$ is α then $2\sqrt{3}\left|\cos\alpha\right|$ =

75. If $f\left(x\right)=\left(2x-3\pi\right)^5+\dfrac{4}{3}x+\cos x$ and g is the inverse function of f, then $g'\left(2\pi\right)$ is equal to 'k' then 7k =